



PRA: Quantitative Project Risk Analysis in R

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Abstract

The **PRA** package for R provides a unified, open-source implementation of the core analytical methods used in quantitative project risk analysis: the Second Moment Method (SMM), Monte Carlo simulation (MCS), earned value management (EVM), contingency and sensitivity analysis, Bayesian risk inference and updating, sigmoidal learning curves, and design structure matrices (DSM). Building on prior theoretical work that introduced causal networks and structural network measures for project risk (Govan 2014; Govan and Damnjanovic 2016, 2020), **PRA** operationalizes these frameworks in a minimal, well-tested R package available on CRAN. Version 0.4.0 adds an AI agent layer that exposes all analytical functions as tools callable by large language models via slash commands, natural language queries, and a retrieval-augmented generation knowledge base. The package adheres to rOpenSci Statistical Software Review standards and is supported by a 640-test suite running on continuous integration across macOS, Windows, and Ubuntu.

Keywords: project risk analysis, Monte Carlo simulation, earned value management, Bayesian inference, design structure matrix, R.

1. Introduction

Quantitative risk analysis is a foundational discipline in project management, enabling organizations to assess schedule and cost uncertainty, identify high-impact risk drivers, and set evidence-based contingency reserves. Govan (2014) and Govan and Damnjanovic (2016) introduced the first implementation of causal networks for project risk analysis grounded in the resource-based view, demonstrating that organizational capabilities and shared resources—not just schedule uncertainty—are the structural pathways through which risks propagate across project tasks. Extending this work, Govan and Damnjanovic (2020) developed structural network measures that quantify the topology of project risk networks, enabling systematic identification of high-centrality risk nodes.

Despite broad industry adoption of quantitative methods, open-source tooling for integrated

project risk analysis in R remains fragmented. Proprietary platforms such as Oracle Primavera Risk Analysis and Palisade @RISK provide integrated workflows but impose licensing costs and limit reproducibility. Within R, no single package unifies Monte Carlo schedule risk, earned value management, Bayesian risk inference, sigmoidal learning curves, and dependency structure analysis under a common API. The **mc2d** package (Pouillot and Delignette-Muller 2010) provides general-purpose two-dimensional Monte Carlo simulation but has no project-specific constructs. The **sensitivity** package implements variance-based indices for general functions but does not integrate with schedule or cost models. EVM metrics are absent from CRAN entirely. Bayesian packages such as **rstan** and **brms** operate at a high level of generality and require substantial expertise to apply in project risk contexts.

PRA fills this gap. It provides a coherent, CRAN-validated framework of ten analytical modules covering the full quantitative project risk workflow—from uncertainty characterization and simulation to performance monitoring and Bayesian updating—while operationalizing the causal and network-theoretic frameworks of Govan (2014) and Govan and Damjanovic (2016, 2020) computationally. The package targets project analysts who work in R and require reproducible, auditable outputs compatible with standard reporting pipelines. It adheres to rOpenSci Statistical Software Review (SRR) standards (rOpenSci Statistical Software Peer Review 2021) and uses minimal hard dependencies: **mc2d** (Pouillot and Delignette-Muller 2010) for triangular distribution families and **minpack.lm** (Elzhov, Mullen, Spiess, and Bolker 2016) for nonlinear least squares fitting. All other capabilities (AI agent, Shiny app, MCP server) activate through optional **Suggests**-only dependencies.

The remainder of this article describes each analytical module in turn, with working code examples drawn from the same R session. Section~2 covers schedule and cost uncertainty methods (SMM, MCS, contingency, sensitivity). Section~3 describes earned value management. Section~4 covers Bayesian risk inference and updating. Section~5 describes sigmoidal learning curves. Section~6 covers design structure matrices and probabilistic networks. Section~7 briefly introduces the AI agent framework. Section~8 presents an integrated case study. Section~9 concludes.

2. Schedule and Cost Uncertainty

Two complementary methods are provided for quantifying schedule and cost uncertainty: the Second Moment Method for rapid analytical estimates, and Monte Carlo simulation for full distributional analysis. Contingency extraction and sensitivity analysis complete the workflow.

2.1. Second Moment Method

The Second Moment Method (SMM) propagates task-level means and variances to project-level estimates using the Central Limit Theorem (Benjamin and Cornell 2000). For a project with n tasks:

$$E[X] = \sum_{i=1}^n E[X_i], \quad \text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \rho_{ij} \sigma_i \sigma_j$$

The **smm()** function takes vectors of task means and variances plus a correlation matrix and returns the total mean, variance, and standard deviation.

```
R> task_means <- c(10, 15, 20) # weeks
R> task_vars <- c(4, 9, 16)
R> cor_mat <- matrix(c(
+   1.0, 0.5, 0.3,
+   0.5, 1.0, 0.4,
+   0.3, 0.4, 1.0
+ ), nrow = 3, byrow = TRUE)
R>
R> result <- smm(task_means, task_vars, cor_mat)
R> cat("Total mean:", round(result$total_mean, 2), "weeks\n")
```

Total mean: 45 weeks

```
R> cat("Total SD: ", round(result$total_std, 2), "weeks\n")
```

Total SD: 7.03 weeks

```
R> ci <- result$total_mean + c(-1, 1) * 1.96 * result$total_std
R> cat("95% CI:      [", round(ci[1], 1), ",", round(ci[2], 1), "] weeks\n")
```

95% CI: [31.2 , 58.8] weeks

SMM is useful when only first and second moments are available and a rapid estimate is needed. When task distributions are non-normal or accurate tail behavior is required, Monte Carlo simulation is preferred.

2.2. Monte Carlo Simulation

`mcs()` propagates uncertainty through a project network by drawing samples from user-specified task distributions—triangular, normal, or uniform—over a large number of iterations (Vose 2008). Task correlations are incorporated via Cholesky decomposition of the supplied correlation matrix.

```
R> num_simulations <- 10000
R> task_distributions <- list(
+   list(type = "normal", mean = 10, sd = 2),
+   list(type = "triangular", a = 5, b = 10, c = 15),
+   list(type = "uniform", min = 8, max = 12)
+ )
R> results <- mcs(num_simulations, task_distributions, cor_mat)
R> cat("Simulated mean:", round(results$total_mean, 2), "weeks\n")
```

Simulated mean: 38.6 weeks

```
R> cat("Simulated SD: ", round(results$total_sd, 2), "weeks\n")
```

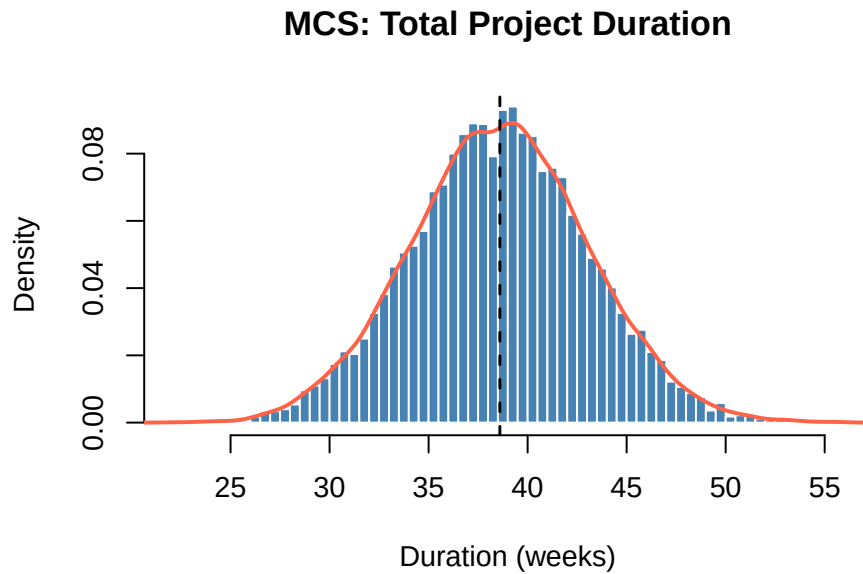


Figure 1: Distribution of simulated total project durations. The dashed vertical line marks the mean.

Simulated SD: 4.47 weeks

The `mcs()` function returns an S3 object of class "mcs" with a `print()` method that summarizes percentiles. The full simulated distribution is available in `results$total_distribution`.

```
R> hist(results$total_distribution,
+       breaks = 50, freq = FALSE,
+       main = "MCS: Total Project Duration",
+       xlab = "Duration (weeks)", col = "steelblue", border = "white"
+ )
R> lines(density(results$total_distribution), col = "tomato", lwd = 2)
R> abline(v = results$total_mean, lty = 2, lwd = 1.5)
```

2.3. Contingency Analysis

`contingency()` extracts a reserve buffer as the difference between a high-confidence percentile and the base (median) estimate—a standard approach in project risk practice ([Project Management Institute 2021](#)).

```
R> reserve <- contingency(results, phigh = 0.80, pbase = 0.50)
R> cat("Schedule contingency (P80 - P50):", round(reserve, 2), "weeks\n")
```

Schedule contingency (P80 - P50): 3.7 weeks

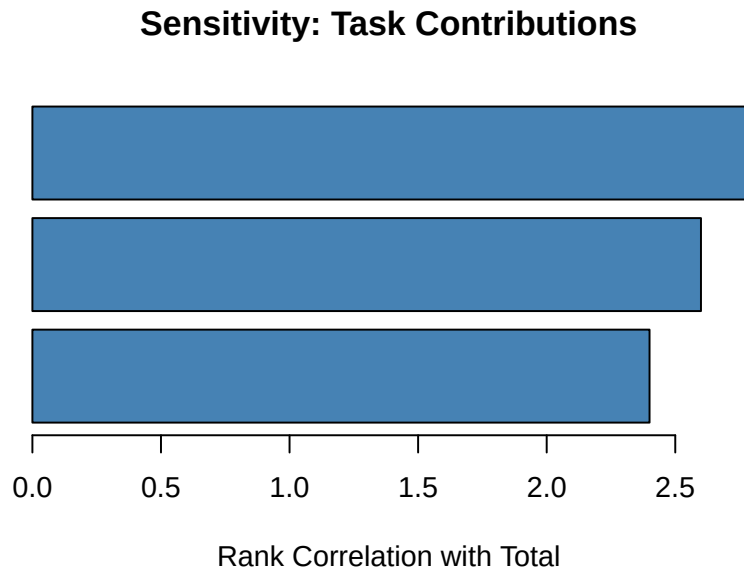


Figure 2: Tornado chart: each bar shows a task's rank correlation with total project duration.

2.4. Sensitivity Analysis

`sensitivity()` decomposes total project variance by task contribution using rank correlation, producing the inputs for a Tornado chart.

```
R> sens <- sensitivity(task_distributions, cor_mat)
R> barplot(sort(sens),
+   horiz = TRUE, col = "steelblue",
+   main = "Sensitivity: Task Contributions",
+   xlab = "Rank Correlation with Total"
+ )
```

Tasks with larger bars are the dominant drivers of total risk; mitigation resources should be directed there first.

3. Earned Value Management

The EVM module implements the complete ANSI/EIA-748 suite of eleven performance metrics (Fleming and Koppelman 2010). The three core quantities—Planned Value (PV), Earned Value (EV), and Actual Cost (AC)—are computed first; all performance indices and forecasts follow from them.

```
R> bac <- 500000 # budget at completion ($)
R> schedule <- c(0.10, 0.25, 0.50, 0.75, 1.00) # cumulative value curve
R> time_period <- 3
```

```
R> pv_val <- pv(bac, schedule, time_period)
R> ev_val <- ev(bac, actual_per_complete = 0.40)
R> ac_val <- ac(c(45000, 110000, 135000), time_period, cumulative = FALSE)
R>
R> cat("PV: $", format(pv_val, big.mark = ","), "\n")
```

PV: \$ 250,000

```
R> cat("EV: $", format(ev_val, big.mark = ","), "\n")
```

EV: \$ 2e+05

```
R> cat("AC: $", format(ac_val, big.mark = ","), "\n")
```

AC: \$ 290,000

```
R> spi_val <- spi(ev_val, pv_val)
R> cpi_val <- cpi(ev_val, ac_val)
R> sv_val <- sv(ev_val, pv_val)
R> cv_val <- cv(ev_val, ac_val)
R>
R> cat("SPI:", round(spi_val, 3), " CPI:", round(cpi_val, 3), "\n")
```

SPI: 0.8 CPI: 0.69

```
R> cat("SV: $", format(round(sv_val), big.mark = ","),
+      " CV: $", format(round(cv_val), big.mark = ","), "\n")
```

SV: \$ -50,000 CV: \$ -90,000

Both $SPI < 1$ and $CPI < 1$ indicate the project is behind schedule and over budget. The `eac()` function provides three completion forecasts under different efficiency assumptions:

```
R> cat("EAC (typical): $", format(round(eac(bac, method = "typical",
+    cpi = cpi_val))), big.mark = ","), "\n")
```

EAC (typical): \$ 725,000

```
R> cat("EAC (atypical): $", format(round(eac(bac, method = "atypical",
+    ac = ac_val, ev = ev_val))), big.mark = ","), "\n")
```

EAC (atypical): \$ 590,000

```
R> cat("EAC (combined): $", format(round(eac(bac, method = "combined",
+    cpi = cpi_val, ac = ac_val, ev = ev_val, spi = spi_val))),
+      big.mark = ","), "\n")
```

EAC (combined): \$ 833,750

The remaining EVM metrics are:

```
R> eac_typ <- eac(bac, method = "typical", cpi = cpi_val)
R> cat("ETC:  $", format(round(etc(bac, ev_val, cpi = cpi_val)), big.mark = ","), "\n")
```

ETC: \$ 435,000

```
R> cat("TCPI:", round(tcpi(bac, ev_val, ac_val), 3), "\n")
```

TCPI: 1.429

```
R> cat("VAC:  $", format(round(vac(bac, eac_typ)), big.mark = ","), "\n")
```

VAC: \$ -225,000

The To-Complete Performance Index (TCPI) reports the cost efficiency required on all remaining work to finish within the original budget; values substantially above 1.0 signal an unrealistic recovery target.

4. Bayesian Risk Methods

Building on the causal structure introduced in Govan (2014) and Govan and Damnjanovic (2016), the Bayesian module models risk event probabilities from root causes and supports formal updating as project observations become available (Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin 2013; Pearl 2009).

4.1. Prior Risk Probability

`risk_prob()` computes the marginal probability of a risk event R from multiple independent root causes using the total probability theorem:

$$P(R) = \sum_i P(R \mid C_i) P(C_i) + P(R \mid \bar{C}_i) P(\bar{C}_i)$$

```
R> cause_probs          <- c(0.30, 0.20)
R> risks_given_causes   <- c(0.80, 0.60)
R> risks_given_not_causes <- c(0.20, 0.40)
R>
R> prior <- risk_prob(cause_probs, risks_given_causes, risks_given_not_causes)
R> cat("Prior P(R):", round(prior, 3), "\n")
```

Prior P(R): 0.82

4.2. Bayesian Updating

`risk_post_prob()` updates the risk probability after observing cause status on site. Unobserved causes are passed as NA and are excluded from the update.

```
R> observed_causes <- c(1, NA) # Cause 1 present; Cause 2 not yet assessed
R> post <- risk_post_prob(cause_probs, risks_given_causes,
+                        risks_given_not_causes, observed_causes)
R> cat("Posterior P(R):", round(post, 3), "\n")
```

Posterior P(R): 0.632

```
R> cat("Change:      ", round(post - prior, 3), "\n")
```

Change: -0.188

Observing Cause~1 (which has a strong link to R) raises the risk probability substantially. The posterior informs contingency updates mid-project.

4.3. Cost Risk Distributions

`cost_pdf()` and `cost_post_pdf()` sample prior and posterior cost distributions by simulating which risks materialize and summing their cost impacts.

```
R> risk_probs      <- c(0.3, 0.5, 0.2)
R> means_given_risks <- c(10000, 15000, 5000)
R> sds_given_risks  <- c(2000, 1000, 1000)
R> base_cost       <- 2000
R>
R> prior_costs <- cost_pdf(5000, risk_probs, means_given_risks,
+                        sds_given_risks, base_cost)
R> post_costs  <- cost_post_pdf(5000,
+ observed_risks = c(1, NA, 0), # Risk 1 present, Risk 2 unobserved, Risk 3 absent
+ means_given_risks, sds_given_risks, base_cost
+ )
R>
R> plot(density(prior_costs), col = "steelblue", lwd = 2,
+      main = "Cost Risk: Prior vs. Posterior",
+      xlab = "Cost ($)", xlim = range(c(prior_costs, post_costs)))
R> lines(density(post_costs), col = "tomato", lwd = 2)
R> legend("topright", legend = c("Prior", "Posterior"),
+      col = c("steelblue", "tomato"), lwd = 2, bty = "n")
```

4.4. Correlation Matrix Construction

`cor_matrix()` constructs a valid positive-definite correlation matrix from a vector of upper-triangle entries, for use as input to `mcs()`.

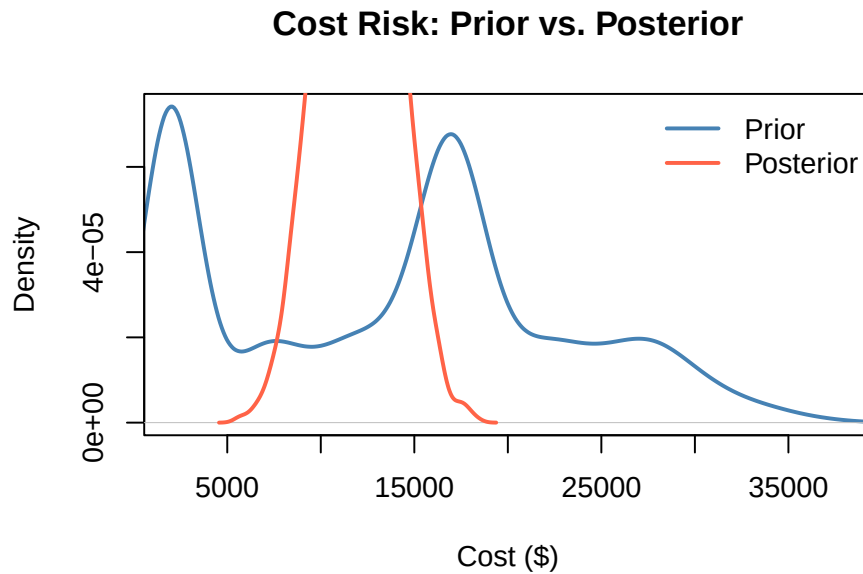


Figure 3: Prior and posterior cost distributions. Observing Cause 1 shifts the distribution rightward, increasing the expected cost impact.

```
R> dists <- list(
+   normal = function(n) rnorm(n, mean = 10, sd = 2),
+   triang = function(n) mc2d::rpert(n, min = 5, mode = 10, max = 15),
+   uniform = function(n) runif(n, min = 8, max = 12)
+ )
R> C_emp <- cor_matrix(num_samples = 1000, num_vars = 3, dists = dists)
R> print(round(C_emp, 3))

      [,1] [,2] [,3]
[1,] 1.000 -0.046 0.015
[2,] -0.046 1.000 -0.005
[3,] 0.015 -0.005 1.000
```

5. Learning Curves

Sigmoidal (S-curve) models capture the characteristic pattern of project progress: slow start, rapid acceleration, and plateau approaching 100% completion. **PRA** provides three model types—Logistic, Pearl, and Gompertz—fitted via the Levenberg–Marquardt nonlinear least-squares algorithm in **minpack.lm** (Elzhov *et al.* 2016).

```
R> progress <- data.frame(
+   time      = 1:9,
+   completion = c(5, 15, 40, 60, 70, 75, 80, 85, 90)
+ )
```

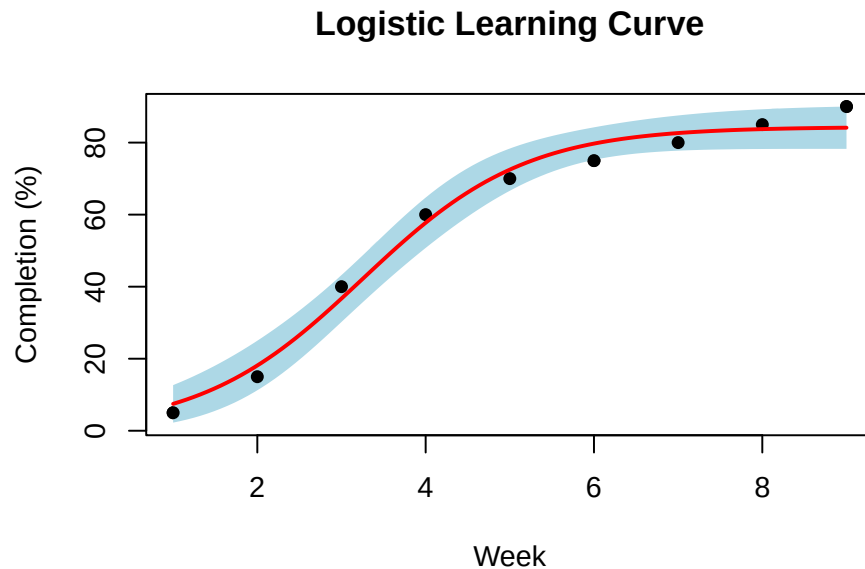


Figure 4: Logistic learning curve with 95% confidence band. The shaded region widens as extrapolation extends beyond the observed data range.

```
R> fit_log <- fit_sigmodal(progress, "time", "completion", "logistic")
R> fit_gomp <- fit_sigmodal(progress, "time", "completion", "gompertz")
R>
R> cat("Logistic RSE: ", round(summary(fit_log)$sigma, 2), "\n")
```

Logistic RSE: 4.15

```
R> cat("Gompertz RSE: ", round(summary(fit_gomp)$sigma, 2), "\n")
```

Gompertz RSE: 2.89

```
R> plot_sigmodal(fit_log, progress, "time", "completion", "logistic",
+               conf_level = 0.95,
+               main = "Logistic Learning Curve",
+               xlab = "Week", ylab = "Completion (%)")
```

`predict_sigmodal()` generates forecasts with confidence bounds at arbitrary future time points:

```
R> preds <- predict_sigmodal(fit_log, seq(10, 12, by = 0.5),
+                           "logistic", conf_level = 0.95)
R> print(round(preds, 1))
```

	x	pred	lwr	upr
1	10.0	84.3	78.2	90.3

```

2 10.5 84.3 78.2 90.4
3 11.0 84.3 78.2 90.4
4 11.5 84.3 78.2 90.5
5 12.0 84.3 78.2 90.5

```

6. Dependency Structure Analysis

Building on Govan and Damnjanovic (2016, 2020) and the foundational DSM literature ([Steward 1981](#); [Browning 2001](#)), the DSM module quantifies structural coupling between project tasks through shared resources and risks.

6.1. Design Structure Matrices

The **Parent DSM** is derived from the Resource–Task matrix $\mathbf{S}$ (rows = resources, columns = tasks) as $\mathbf{P} = \mathbf{S}^\top \mathbf{S}$. Off-diagonal entry P_{jk} counts the number of resources shared between tasks j and k —the structural pathway through which disruptions propagate ([Govan and Damnjanovic 2016](#)).

```

R> S <- matrix(c(
+   1, 0, 1, 0,
+   1, 1, 0, 0,
+   0, 1, 0, 1,
+   0, 0, 1, 1,
+   0, 1, 1, 0
+ ), nrow = 4, ncol = 5)
R> rownames(S) <- paste0("R", 1:4)
R> colnames(S) <- paste0("T", 1:5)

R> p <- parent_dsm(S)
R> plot(p)

```

The **Grandparent DSM** adds a risk layer via the Risk–Resource matrix $\mathbf{R}$, tracing dependencies from risks through resources to tasks. Off-diagonal entry G_{jk} counts the number of risks shared between task j and task k through the resource chain—a direct operationalization of the structural network measures developed in Govan and Damnjanovic (2020).

```

R> R_mat <- matrix(c(
+   1, 0, 1,
+   1, 1, 0,
+   0, 1, 0,
+   0, 0, 1
+ ), nrow = 3, ncol = 4)
R> rownames(R_mat) <- paste0("Risk", 1:3)
R> colnames(R_mat) <- paste0("R", 1:4)

R> g <- grandparent_dsm(S, R_mat)
R> plot(g)

```

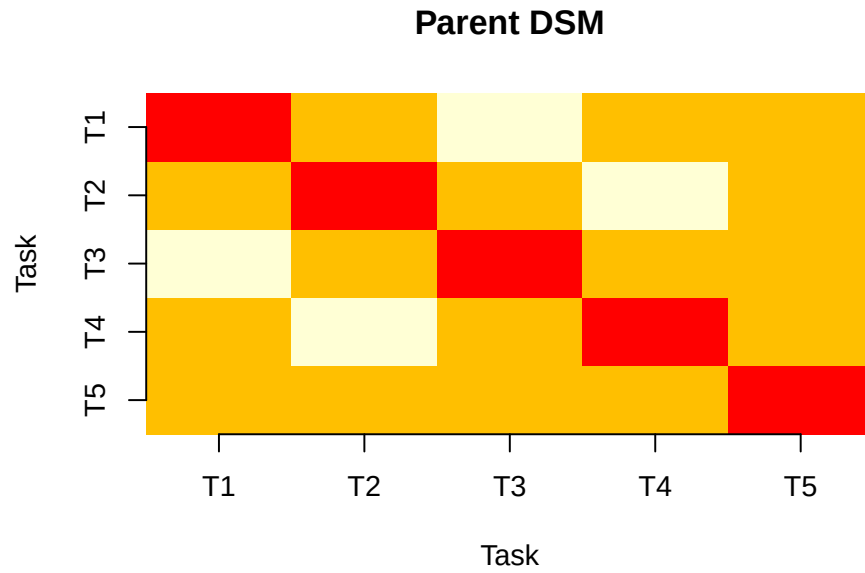


Figure 5: Parent DSM heatmap. Darker cells indicate more shared resources and stronger coupling between task pairs.

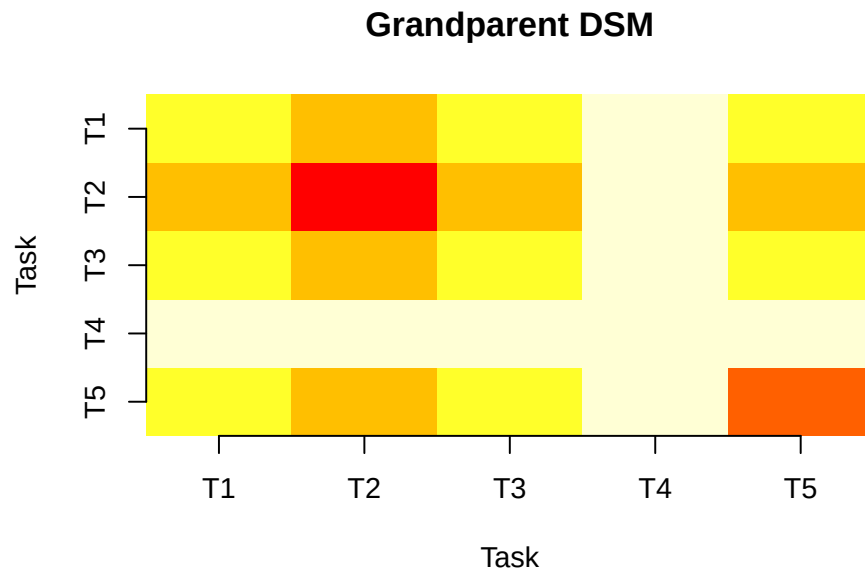


Figure 6: Grandparent DSM heatmap. Shows risk-based coupling: task pairs that share risk exposure through common resources tend to fail together.

Task pairs with high off-diagonal values in the Grandparent DSM represent concentrated risk exposure; these pairs are prime candidates for contingency buffering or resource decoupling.

7. AI Agent Framework

Version 0.4.0 of **PRA** includes an optional AI agent layer that wraps all analytical functions as tools callable by large language models. Three interaction modes are supported:

1. **Slash commands** (`/mcs`, `/evm`, `/risk`, etc.) for deterministic, LLM-free execution via `PRA:::execute_command()`.
2. **Natural language** through `pra_chat()`, which connects to a local Ollama model (default: `llama3.2`) and uses **ellmer** (Wickham *et al.* 2025) tool-calling to route queries to the appropriate analytical function.
3. **Retrieval-augmented generation (RAG)** via **ragnar** (Posit PBC 2025): conceptual questions are answered from a bundled domain knowledge base with hybrid BM25 + vector search and source attribution.

An interactive Shiny application (`pra_app()`) integrates all three modes with a chat interface, data upload, and inline plots. `pra_mcp_server()` exposes all tools via the Model Context Protocol (Anthropic 2024), enabling integration with Claude, Claude Code, and other MCP-compatible clients.

The AI layer is strictly optional: the 36 exported analytical functions operate without any AI dependency. Tool definitions built with `ellmer::tool()` are reused identically for both the chat interface and the MCP server, eliminating duplication. A preliminary evaluation of 25 scenarios across single-tool, multi-tool, and open-ended query tiers yielded 72.0% overall accuracy (87.5% single-tool, 55.6% multi-tool, 75.0% open-ended), with the RAG layer providing the largest accuracy gain on open-ended conceptual queries. A full empirical evaluation is the subject of a forthcoming publication in *Risk Analysis*.

8. Illustrative Case Study

This section demonstrates an integrated, end-to-end project risk analysis for a three-task construction project, progressing through uncertainty quantification, performance monitoring, Bayesian risk updating, and dependency structure analysis in a single R session.

8.1. Step 1: Characterize Task Uncertainty

```
R> tasks <- list(
+   list(type = "triangular", a = 8, b = 12, c = 18), # earthworks
+   list(type = "normal", mean = 20, sd = 3), # structural frame
+   list(type = "triangular", a = 5, b = 8, c = 12) # MEP
+ )
R> C_proj <- matrix(c(1, 0.4, 0.2,
```

```

+           0.4, 1, 0.3,
+           0.2, 0.3, 1), nrow = 3, byrow = TRUE)
R>
R> sim <- mcs(10000, tasks, C_proj)
R> res <- contingency(sim, phigh = 0.80, pbase = 0.50)
R> cat("P50 schedule:", round(sim$percentiles["50%"], 1), "weeks\n")

P50 schedule: 51.3 weeks

R> cat("P80 schedule:", round(sim$percentiles["80%"], 1), "weeks\n")

P80 schedule: NA weeks

R> cat("Contingency (P80-P50):", round(res, 2), "weeks\n")

Contingency (P80-P50): 4.31 weeks

```

8.2. Step 2: Monitor Earned Value at Mid-Project

```

R> bac_cs <- 1200000
R> sched_cs <- c(0.15, 0.35, 0.60, 0.85, 1.00)
R> pv_cs <- pv(bac_cs, sched_cs, time_period = 3)
R> ev_cs <- ev(bac_cs, actual_per_complete = 0.50)
R> ac_cs <- ac(c(180000, 250000, 310000), time_period = 3, cumulative = FALSE)
R>
R> cat("SPI:", round(spi(ev_cs, pv_cs), 3),
+      " CPI:", round(cpi(ev_cs, ac_cs), 3), "\n")

SPI: 0.833 CPI: 0.811

R> cat("EAC (typical): $", format(round(eac(bac_cs,
+      method = "typical", cpi = cpi(ev_cs, ac_cs))), big.mark = ","), "\n")

EAC (typical): $ 1,480,000

```

8.3. Step 3: Update Risk Probability After Site Inspection

```

R> prior_r <- risk_prob(c(0.25, 0.15), c(0.70, 0.50), c(0.15, 0.30))
R> post_r <- risk_post_prob(c(0.25, 0.15), c(0.70, 0.50),
+                           c(0.15, 0.30), observed_causes = c(1, NA))
R> cat("Prior P(R):", round(prior_r, 3),
+      "-> Posterior P(R):", round(post_r, 3), "\n")

Prior P(R): 0.617 -> Posterior P(R): 0.609

```

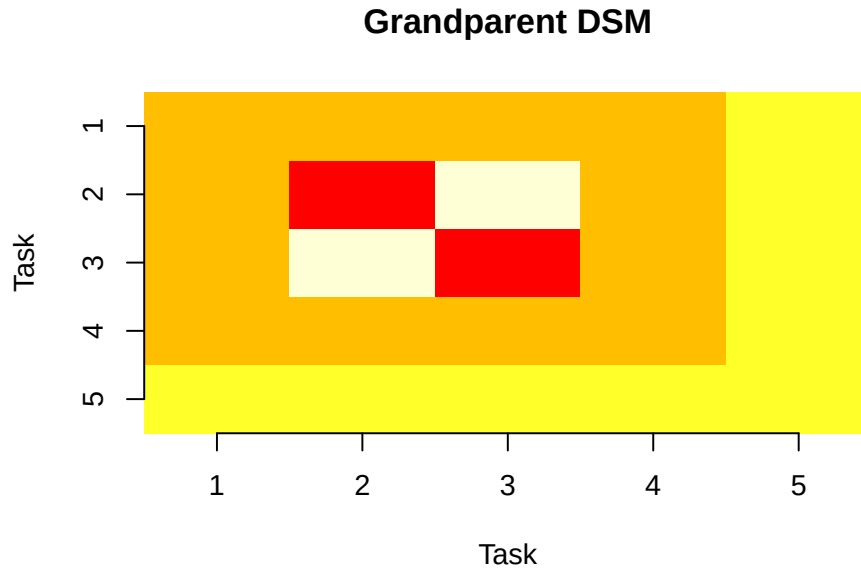


Figure 7: Grandparent DSM for the case-study project. Task pairs T3–T5 share the most risk exposure through common resources.

8.4. Step 4: Identify Coupled Tasks via DSM

```
R> S_cs <- matrix(c(
+   1, 1, 0, 0,
+   0, 1, 1, 0,
+   1, 0, 0, 1,
+   0, 0, 1, 1,
+   0, 1, 0, 1
+ ), nrow = 4, ncol = 5)
R> R_cs <- matrix(c(
+   1, 0, 1,
+   0, 1, 0,
+   1, 1, 0,
+   0, 0, 1
+ ), nrow = 3, ncol = 4)
R> g_cs <- grandparent_dsm(S_cs, R_cs)
R> plot(g_cs)
```

The Grandparent DSM reveals which task pairs share risk exposure through common resources, informing targeted contingency allocation and resource decoupling decisions.

9. Summary

PRA is a CRAN-available R package that provides a unified, open-source implementation of quantitative project risk analysis methods. It operationalizes the causal network and structural network frameworks developed in Govan (2014) and Govan and Damnjanovic (2016, 2020) through ten integrated analytical modules: SMM, MCS, contingency analysis, sensitivity analysis, EVM (11 metrics), Bayesian risk inference and updating, cost risk distributions, sigmoidal learning curves, correlation matrix construction, and design structure matrices.

Version 0.4.0 adds an AI agent layer that exposes all functions as tools callable via slash commands, natural language, and RAG-powered queries through a local Shiny application and an MCP server. All 36 exported functions adhere to rOpenSci SRR standards, with 640+ tests verifying correctness, edge-case handling, and numerical stability across three operating systems. The package is licensed under MIT and the source is available at <https://github.com/paulgovan/PRA>; documentation and a companion online book are at <https://paulgovan.github.io/PRA/>.

A forthcoming companion article in *Risk Analysis* presents a full empirical evaluation of the AI agent framework, including a 25-scenario benchmark, RAG retrieval quality analysis, and a comparison to purely LLM-based assistants.

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